For Examiner's Use

3 **(a)** State Archimedes' Principle.

……

……

…….…….... [1]

(b) Rectangular wooden blocks are used in a water game where participants skip from one block to another. One such block of uniform mass having a cross-sectional area of 0.54 m^2 and thickness 30 cm floats in sea water with 20 cm submerged in water as shown in Fig. 3.1. The density of sea water is 1.1×10^3 kg m^{-3}.

Two forces, F_T and F_B, act on the top and bottom surfaces of the block respectively due to the pressure exerted by the air and water. The atmospheric pressure above the surface of the water is 1.0×10^5 Pa.

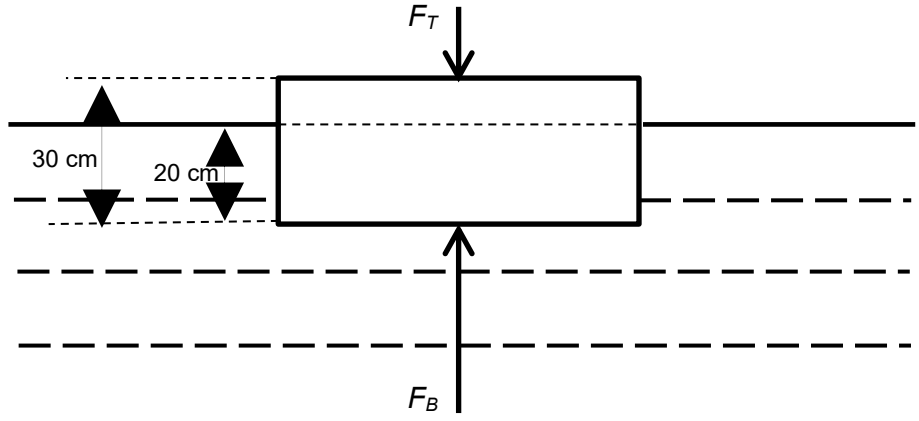

Fig. 3.1

(i) Show that the resultant force F_R acting on the block due to F_T and F_B is 1200 N.

[1]

(ii) On Fig. 3.1, draw a single arrow to represent F_R. [1]

(iii) Determine the density of the wooden block.

density = …… kg m^{-3} [2]

For Examiner's Use

(iv) A participant steps on a block and causes it to be submerged to a depth of 10 cm as shown in Fig. 3.2.

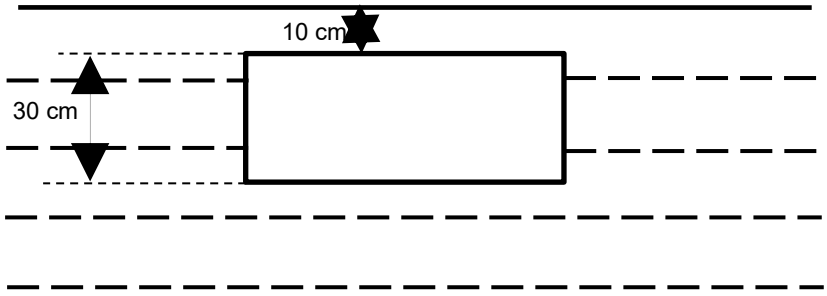

Fig. 3.2

The participant then steps off the block.

1. Determine the acceleration *a* acting on the block.

a = …… m s^{-2} [2]

2. Hence, calculate the time taken for the top surface of the block to just reach the surface of the water. Assume that acceleration is constant.

time = …… s [2]

3. The actual time taken is different from that found in **(b)(iv)2**.
State and explain this difference.

……...

……...

……...

…… [2]

[Turn over